EXP-RAG vs Vanilla (AI Studio)

Yuzee AI Token Lab · generated 3 October 2026 at 8:41 pm · model gemini-3.7-flash (medium thinking, both modes) · judge gemini-3.7-flash (blind, pairwise)

12–8–16
EXP-RAG · BGE: wins–losses–ties vs Vanilla (36 turns)

15–4–17
EXP-RAG · Gemini: wins–losses–ties vs Vanilla (36 turns)

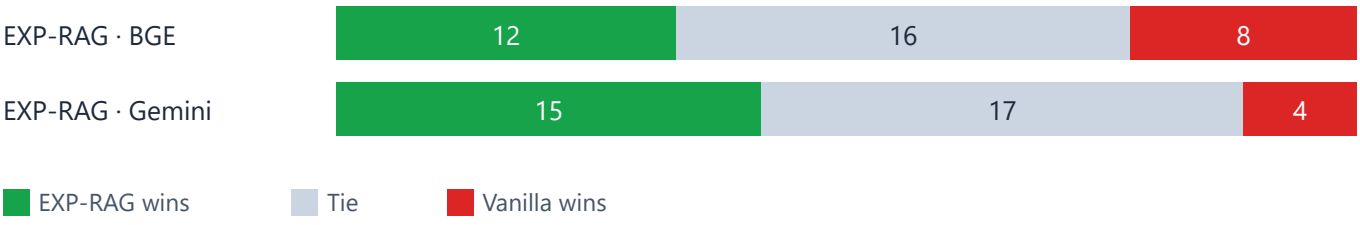
-17%
Median latency, EXP-RAG · Gemini vs Vanilla (14.2 s vs 17.2 s)

-36%
Cost per turn, EXP-RAG · Gemini ($0.0027 vs $0.0042)

Summary

- **EXP-RAG with Gemini embeddings (the default) is significantly better than Vanilla:** 15 wins, 4 losses and 17 ties over 36 turns judged in both answer orders, where a win counts only if both orders agree (sign test p = 0.01). It scores higher on counselling (4.78 → 4.92) and clarity, with equal relevance, grounding and continuity within ±0.06.
- **EXP-RAG with BGE embeddings is on par with Vanilla, not significantly better:** 12–8–16 (p = 0.25), total 24.58 → 24.53. It shares every speed and cost gain, so it is the right choice where no Gemini API key is available, but Gemini is the better retriever for quality.
- **Faster:** median turn latency 14.2 s for both EXP-RAG variants vs 17.2 s for Vanilla (−17% pooled; −23 to −25% in the run where every mode was measured in the same window). The gain comes from before the model call (3.1 → 2.1–2.3 s), time to first token (4.0 → 3.2 s) and the review (4.4 → 2.7–2.8 s); generation time is about the same.
- **Cheaper:** 36–37% lower cost per turn ($0.0027 vs $0.0042) with about a third of Vanilla's input tokens: the cached core prompt replaces the full prompt and only relevant guidance and evidence are retrieved.
- **Issues found along the way:** comparison tables showed no row labels to users (fixed for every mode); the teaching review re-emitted every substantive answer in full, about 4 s per turn (fast path for EXP-RAG; Vanilla left unchanged so the baseline stays fair); and EXP-RAG wrote a short poem for an off-topic request, which breaks Oala's scope (fixed: it now declines and redirects in 4/4 runs).
- **Judge noise is large:** re-scoring identical answers once moved several verdicts, so the headline uses both final runs, each judged in both orders. Single-order counts are higher (Gemini 23–8–5) but less trustworthy.

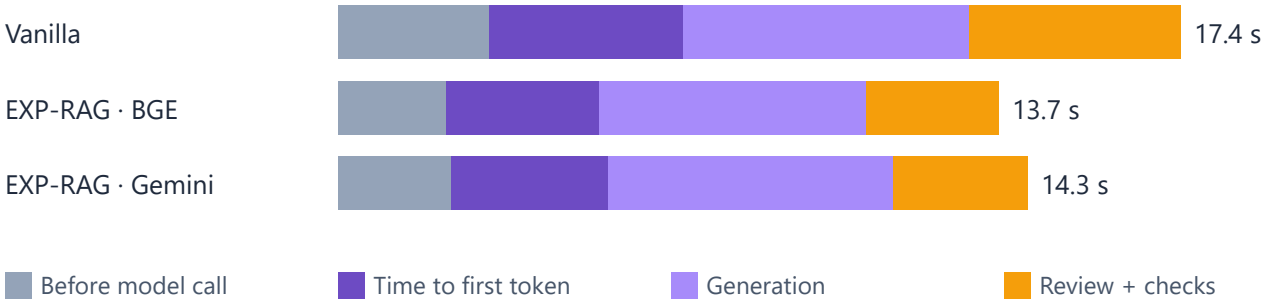
Head-to-head quality



Criterion (1–5)	Vanilla	EXP-RAG · BGE	EXP-RAG · Gemini
Relevance	4.99	4.90	4.94
Grounding	4.99	4.93	4.97
Counselling	4.75	4.82	4.92
Clarity	4.92	4.92	4.96
Continuity	4.99	4.96	4.97
Total (of 25)	24.65	24.53	24.76
Sign test on decisive turns (one-sided p)	—	0.252	0.010

Vanilla scores are averaged across the judge's two pairings. Ties are excluded from the sign test.

Speed and cost



Per turn	Vanilla	EXP-RAG · BGE	EXP-RAG · Gemini
Median latency	17.2 s	14.2 s	14.2 s
Mean latency	17.4 s	13.7 s	14.3 s
Before model call	3.1 s	2.2 s	2.3 s
Time to first token	4.0 s	3.2 s	3.3 s
Generation	5.9 s	5.5 s	5.9 s
Review + checks	4.4 s	2.7 s	2.8 s
Input tokens	65,507	24,004	23,931
…of which cached	46,936	8,739	8,538
Output tokens	2,038	1,453	1,445
Thinking tokens	966	840	926
Cost (USD)	$0.00423	$0.00266	$0.00270
Valid protocol responses	36/36	36/36	36/36

Tuning history

Round	Change tested	Variant	W–L–T	Score V→E	Median latency V→E	Cost/turn V→E
1 · Baseline	EXP-RAG as built (BGE), re-judged with a fair renderer (internal ids hidden)	BGE	5–5–3	23.62 → 24.15	20.0 → 18.5 s (mean)	$0.00546 → $0.00302
2	Concreteness rule; open question preferred over a menu; comparison row labels fixed in the UI	BGE	4–6–3	24.46 → 24.23	20.6 → 16.3 s (mean)	$0.00543 → $0.00296
3	Single-question guard; explicit final check; judge told how tables are shown	BGE	6–3–4	24.62 → 24.62	18.5 → 16.8 s (mean)	$0.00472 → $0.00288
4	4 more scenarios (18 turns); no-restating rule; Gemini embeddings added	BGE	8–5–5	24.50 → 24.61	17.9 → 17.7 s	$0.00497 → $0.00280
4	(same)	Gemini	5–10–3	24.78 → 24.17	17.9 → 19.5 s	$0.00497 → $0.00297
5	Review fast path; Flash-Lite planner; interstate courses dropped; rerank	BGE	7–6–5	24.28 → 24.72	17.9 → 15.6 s	$0.00497 → $0.00288

	skipped when it cannot change the result					
5	(same)	Gemini	10–6–2	24.44 → 24.67	17.9 → 15.0 s	$0.00497 → $0.00275
Final A	Compound-question trim; codes rule. Every mode run fresh in the same window	BGE	10–5–3	24.33 → 24.44	17.3 → 13.3 s	$0.00423 → $0.00255
Final A	(same)	Gemini	14–3–1	24.33 → 24.83	17.3 → 13.0 s	$0.00423 → $0.00255
Final B	Town-level and catalogue local scope; planner comparison guard (vs Final A's Vanilla answers)	BGE	10–6–2	24.44 → 24.44	17.3 → 15.4 s	$0.00423 → $0.00275
Final B	(same)	Gemini	9–5–4	24.39 → 24.61	17.3 → 15.1 s	$0.00423 → $0.00281
Final (robust)	Off-topic scope rule tightened (no poems); both final runs re-judged in both answer orders, win only when both agree	BGE	12–8–16	24.58 → 24.53	17.2 → 14.2 s	$0.00423 → $0.00266
Final (robust)	(same)	Gemini	15–4–17	24.71 → 24.76	17.2 → 14.2 s	$0.00423 → $0.00270

What was changed

Change	Why
Comparison rows labelled from their ids in the UI (all modes)	v1.3 rows carry only id + cells, so the dimension column was hidden from users
Core prompt: concreteness rule (place, role, transferable experience, practical trade-offs as things to check)	Losses for generic advice versus Vanilla's personalised detail
Core prompt: open TEXT question when it reveals more than a menu; explicit one-question final check	Losses for rigid multiple-choice and double questions
Deterministic single-question guard: drops a trailing prose question and trims compound free-text questions	14 of 34 follow-up questions asked two things
Core prompt: no restating table content in prose; codes only when asked and evidenced	Losses for repetition and superseded qualification codes
Teaching review fast path for EXP-RAG: verdict OK or REVISE with blocks	The review re-emitted the full answer every turn (~4 s)
Warehouse planner on Gemini 3.5 Flash-Lite for EXP-RAG, with a comparison guard	Planner latency 2.6 s → 1.3 s; the guard stops a false 'comparison' plan
Local scope for course evidence: interstate-only records dropped, same-state campuses away from the user labelled (lookup and catalogue hits)	Answers named interstate or distant providers as local options
Dense evidence rerank skipped when ranking cannot change the selected set	Saves an embedding round trip (≈0.8 s with Gemini) on most turns
Gemini Embedding 2 as the default EXP-RAG embedder when an API key is set; per-provider thresholds tuned on the guidance eval set	Dense retrieval without local model files; 100% guidance recall with fewer tokens
Off-topic rule: never produce the unrelated content itself	EXP-RAG wrote a short poem for 'write me a poem about cats' in 3 of 4 runs

Evaluation: order-swapped double judging, scope-aware rubric, 429 back-off, UI-faithful rendering	Single-pass judging and rendering artifacts produced misleading verdicts

Per-turn results

Turn	Focus	EXP-RAG · BGE	EXP-RAG · Gemini
`A m1-memory#1`	multi-turn continuity and constraint recall	**EXP-RAG** 24.5/25 Answer B provides more comprehensive pathways and includes a crucial practical counselling note regarding the loss of penalty rates when leaving shift work.	**EXP-RAG** 25/25 Answer B provides excellent, realistic career counselling by proactively highlighting the practical pay trade-off regarding penalty rates when transitioning from shift work to daytime hours.
`A m1-memory#2`	multi-turn continuity and constraint recall	**Vanilla** 25/24 Answer A asks a better open counselling question; Answer B ends with an incomplete/awkwardly phrased question.	Tie 25/24.5 Split verdict (VANILLA / EXP): Answer A maintained continuity with the user's location (Brisbane) and offered a more focused set of pathways and a thoughtful follow-up question.
`A m1-memory#3`	multi-turn continuity and constraint recall	Tie 25/25 Both provide excellent, accurate comparisons tailored to Priya's nursing background and request for regular hours, with clear next-step questions.	**EXP-RAG** 25/25 Both are excellent, but B provides sharper practical detail regarding qualifications (Cert IV TAE40122) and super-user pathways, plus clean question options.
`A m1-memory#4`	multi-turn continuity and constraint recall	Tie 25/25 Both models give accurate, grounded advice comparing study loads and budget constraints for an RN, offering clear structured comparisons and focused follow-up questions.	Tie 25/25 Both models provide exceptionally clear, grounded, and practical comparisons tailored directly to Priya's budget, study constraints, and nursing background.
`A m1-memory#5`	multi-turn continuity and constraint recall	Tie 25/25 Both models accurately recap all stated constraints, give well-reasoned recommendations grounded in the user's situation, and offer practical next steps with appropriate interactive questions.	Tie 25/25 Both models perfectly recall all constraints, accurately compare clinical education against health informatics within budget/study limits, and offer tailored, practical next steps.
`A m2-short#1`	short reply to a pending question	Tie 24.5/24.5 Split verdict (VANILLA / EXP): Answer A asks a better, more tailored interactive question that helps guide the next steps towards study vs practical experience.	**EXP-RAG** 24.5/25 Answer B provides a clearer stepped pathway, suggesting introductory Certificate II/III courses before committing to the placement-heavy Certificate IV.
`A m2-short#2`	short reply to a pending question	Tie 23.5/23.5 Split verdict (EXP / VANILLA): Following a single affirmative response, Answer B appropriately handles the conversation flow by exploring the user's availability and real-life context rather than dumping structured tables.	**Vanilla** 25/23 Answer A provides a clearer structured comparison between practical tasters and formal study, directly addressing the user's lack of experience.
`A s1-explore#1`	first turn, undecided school leaver	**EXP-RAG** 24/25 Answer B adheres strictly to asking at most one focused question, whereas Answer A includes a question in the main text as well as the interactive question.	**EXP-RAG** 24/25 Answer B adheres to asking only one focused question, whereas Answer A asks a question in the body text and another in the prompt tag.
`A s2-compare#1`	qualification comparison	**Vanilla** 25/25 Both answers are excellent. Answer A provides slightly more accurate AQF framing and more realistic initial job outcomes for a Cert IV graduate.	Tie 24.5/25 Split verdict (VANILLA / EXP): Answer A includes AQF levels for precise context and provides a more targeted, actionable closing question to help the user choose between the pathways.

`A s3-cost#1`	fees and funding; must not invent prices	Tie 24.5/23.5 Split verdict (VANILLA / EXP): Answer A is better structured, provides clearer details on practical placement requirements and VSL, and avoids repetitive references to catalogue records.	Tie 24.5/25 Both models give accurate, comprehensive explanations of Victorian nursing subsidies (Skills First), VET Student Loans, and additional placement costs, with well-targeted follow-up questions.
`A s4-change#1`	mid-career change using experience	EXP-RAG 24/25 Both are strong, but B gives slightly sharper practical guidance, mentioning internal secondments and clearly setting expectations around role seniority and salary trade-offs.	Tie 25/25 Split verdict (EXP / VANILLA): Answer B includes specific Australian vocational qualification codes and explicitly clarifies Australian regulatory/employment context for HR roles.
`A s5-outlook#1`	job outlook; forecast honesty	EXP-RAG 23.5/25 Answer B provides a more natural, practical explanation with excellent counselling context on Victorian drivers and entry pathways.	EXP-RAG 23/25 Answer B provides more practical career context, including demand drivers and entry qualifications, written in a clear, natural counselling tone.
`A s6-support#1`	anxiety and disability, supportive accuracy	Tie 25/25 Both answers provide empathetic, realistic comparisons between TAFE and university study styles, explain accessibility supports clearly, and offer a practical next step.	Tie 25/25 Both models provide highly empathetic, accurate, and structured comparisons of TAFE and university, highlighting relevant dyslexia support options.
`A s7-offtopic#1`	off-topic request	Tie 25/25 Both answers correctly and concisely decline the off-scope request and offer an appropriate career-related redirect to animal pathways.	Tie 25/25 Both models correctly decline the out-of-scope creative writing request and smoothly redirect to animal-related career pathways.
`A s8-action#1`	service request beyond capability	Tie 25/25 Both models clearly decline the action request while providing excellent, grounded guidance on pursuing a carpentry apprenticeship in Geelong.	Vanilla 25/23.5 Both correctly decline applying on behalf of the user, but A offers more practical, actionable guidance and a more targeted follow-up question (White Card/licence).
`A s9-rpl#1`	recognition of experience without a degree	EXP-RAG 25/25 Both provide excellent explanations of RPL, but B adds helpful context with specific Australian qualification codes and industry context regarding portfolio value.	EXP-RAG 24.5/25 Answer B provides concrete Australian qualification examples (ICT40120, ICT50220) directly relevant to a web developer seeking RPL.
`A s10-local#1`	local course lookup from warehouse data	Vanilla 25/21 A names the actual standard qualification (Cert III in Individual Support) and focuses strictly on Geelong (The Gordon), whereas B mentions distant TAFE Gippsland and non-standard course titles.	Vanilla 25/24 Answer A provides a more conversational and tailored follow-up question, whereas Answer B's question is rather curt.
`A s11-postgrad#1`	degree vs postgraduate decision with prior degree	Tie 24.5/25 Split verdict (VANILLA / EXP): Both answers are excellent, but A provides a slightly more actionable closing question to help guide the next steps based on current programming experience.	EXP-RAG 24/25 B provides a clearer comparison incorporating AQF levels and finishes with a single, highly focused counselling question rather than a compound one.
`B m1-memory#1`	multi-turn continuity and constraint recall	Tie 25/25 Both provide excellent, empathetic career advice tailored to an Australian RN seeking standard hours, with clear pathways and strong counselling questions.	EXP-RAG 25/25 Answer B provides crucial, practical counselling by highlighting the realistic trade-off of losing shift penalties and weekend loadings when moving away from rotating rosters.
`B m1-memory#2`	multi-turn continuity and	Tie 25/25 Both models give outstanding, highly relevant, realistic	EXP-RAG 25/25 B provides more comprehensive, concrete career

	constraint recall	career transitions for an Australian RN seeking business hours, correctly maintaining continuity and offering thoughtful next steps.	pathways (including Occupational Health) and practical insight regarding the transition from shift penalties.
`B m1-memory#3`	multi-turn continuity and constraint recall	**EXP-RAG** 24/25 Answer B personalises using Priya's name, gives more practical and realistic stepping-stone advice (super-user roles, precepting), and includes helpful multiple-choice options in the question.	**EXP-RAG** 25/25 Both models give outstanding answers, but B provides more concrete entry-level job titles, practical stepping stones (super-user, preceptor), and structured quick-reply options.
`B m1-memory#4`	multi-turn continuity and constraint recall	**EXP-RAG** 24.5/25 Answer B provides a more actionable counselling response with tailored Australian higher education context (CSP funding, Grad Certs) and a natural, practical closing question.	**EXP-RAG** 25/25 B directly mentions FEE-HELP to address the $5,000 budget constraint realistically, providing clearer and more practical Australian education guidance.
`B m1-memory#5`	multi-turn continuity and constraint recall	**Vanilla** 25/24.5 Answer A provides a more practical and targeted follow-up question directly advancing the recommended clinical education pathway.	Tie 25/24.5 Both models perfectly summarise Priya's constraints, accurately evaluate clinical education as the best fit within her budget, and provide excellent, highly grounded counselling.
`B m2-short#1`	short reply to a pending question	**EXP-RAG** 24/25 Answer B provides specific national qualification codes and asks a single, focused closing question, whereas A asks a compound question.	Tie 24.5/25 Both answers give excellent, practical pathways for entering vet nursing with zero experience, outlining Australian qualifications accurately alongside helpful guidance questions.
`B m2-short#2`	short reply to a pending question	**Vanilla** 25/21.5 Answer A provides a practical, informative comparison directly addressing how to enter vet nursing without prior animal experience.	Tie 25/25 Both answers provide clear, well-grounded pathways comparing direct vocational study with entry-level volunteering, accompanied by relevant next-step questions.
`B s1-explore#1`	first turn, undecided school leaver	**EXP-RAG** 23.5/25 Answer A includes two closing questions (one in the text and one as [Question]), whereas Answer B keeps strictly to one focused question.	**EXP-RAG** 24/25 Answer B adheres strictly to the single focused question rule, whereas Answer A includes an extra competing question in the main text.
`B s2-compare#1`	qualification comparison	**Vanilla** 25/24.5 Answer A provides slightly more realistic starting expectations for Cert IV graduates, noting that dedicated SOC roles typically align more naturally with Bachelor/Diploma/experience.	Tie 25/25 Both models provide exceptionally clear, grounded, and realistic comparisons of the AQF levels, realistic job outcomes, and practical trade-offs with well-targeted follow-up questions.
`B s3-cost#1`	fees and funding; must not invent prices	**EXP-RAG** 24/25 Answer B asks a single, tightly focused question about residency to assess funding eligibility, whereas Answer A asks a compound question.	**EXP-RAG** 24/25 Answer B asks a single, highly relevant diagnostic question about prior qualifications, whereas Answer A asks a multi-part question.
`B s4-change#1`	mid-career change using experience	**Vanilla** 25/24.5 Answer A provides a more structured breakdown combining RPL with study options and asks a more helpful, contextualised counselling question.	**EXP-RAG** 25/25 Answer B provides superior career counselling by realistically highlighting the salary/seniority trade-off when transitioning from store management to entry HR roles.
`B s5-outlook#1`	job outlook; forecast honesty	**EXP-RAG** 23.5/25 Answer B is clearer and avoids the repetitive restatement of percentage figures found in Answer A, while providing helpful pathway context.	**EXP-RAG** 23/25 Answer B is cleaner and better structured without the repetitive summary points found in Answer A.

`B s6-support#1`	anxiety and disability, supportive accuracy	Tie 25/25 Both models give outstanding, empathetic, and accurate Australian career counselling, clearly distinguishing TAFE and university assessment models while highlighting disability support.	Tie 25/25 Both models provide excellent, reassuring, and accurate Australian career counselling, explaining the differences between TAFE and university while detailing dyslexia support services.
`B s7-offtopic#1`	off-topic request	Tie 25/25 Both models appropriately decline the out-of-scope request and offer an excellent, constructive redirect to animal-related career pathways.	Tie 25/25 Both correctly decline the out-of-scope creative writing request and pivot effectively to relevant animal-related career pathways.
`B s8-action#1`	service request beyond capability	Tie 24.5/25 Both models clearly decline the action they cannot perform, provide accurate Geelong-specific pathways, and ask a relevant follow-up question.	Tie 25/25 Both models clearly decline the impossible action, provide accurate local avenues in Geelong, and ask an appropriate single follow-up question.
`B s9-rpl#1`	recognition of experience without a degree	Tie 25/25 Both models accurately explain RPL for a self-taught developer, provide realistic evidence examples, and ask a relevant clarifying question about career motivations.	Tie 25/25 Both models give accurate, well-grounded, and structured explanations of RPL in Australia with clear next steps and focused follow-up questions.
`B s10-local#1`	local course lookup from warehouse data	Vanilla 25/21.5 Answer A correctly identifies the standard entry qualifications (Certificate III and IV in Individual Support/Disability) rather than skipping straight to skill sets and a diploma.	Vanilla 25/22 A directly identifies the standard TAFE disability support qualifications without drifting into unrequested university degrees or generic duration estimates like B.
`B s11-postgrad#1`	degree vs postgraduate decision with prior degree	EXP-RAG 24/25 Answer B provides a more comprehensive, nuanced comparison of the career leverage and trade-offs, along with a cleaner single-focus question.	Tie 25/25 Both models give outstanding, well-grounded comparisons tailored to an engineering background with clear tables and effective follow-up questions.

Method

- 36 turns from 26 scenarios (single-turn and multi-turn), sent to the live server with identical model, thinking level, warehouse data, Neo4j/TimesFM intelligence and review pipeline. Only the prompt/RAG assembly differs.
- Each EXP-RAG answer is judged against the Vanilla answer to the same turn by gemini-3.7-flash at temperature 0 with a fixed rubric (relevance, grounding, counselling, clarity, continuity). Answer order is alternated to cancel position bias; the judge is told today's date and how flattened tables are shown.
- Answers are judged as users see them: protocol JSON rendered to visible text, internal ids excluded, comparison rows labelled.
- Evaluation chats ran on an isolated server instance (no shared database).

Limitations

- One generation per turn per run (two final runs pooled) and an LLM judge. Judging both orders and counting only agreed wins removes position bias but leaves many ties; a larger scenario set would be needed to show a significant BGE difference.
- The judge (Gemini 3.7 Flash) is the same model family as the generator; it was blind to which mode produced each answer and saw answers in alternating order.
- 18 turns across 13 scenarios cover first turns, comparisons, costs, outlook, support needs, off-topic and action requests, recognition of experience, local lookups, postgraduate choices and two multi-turn conversations; production traffic will be broader.
- Final B reused Final A's Vanilla answers (Vanilla code was unchanged), so its latency comparison spans two time windows.
- TimesFM forecasts were historical-only during testing (the JSA refresh was unreachable), so no turn exercised a current forecast.

Reproduce: `npm run eval:modes` then `npm run report:modes`. Raw answers and scores: `docs/reports/data/exp-rag-vs-vanilla.json`.